

Process Documentation

Platos Health – Data Analyst Technical Assessment

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Objective

The goal of this analysis was to understand how users interact with the platform and how that behaviour relates to health outcomes.

The work focused on three main areas:

1. Understanding where users drop off during the signup process.
2. Measuring how well users return to the platform after signing up.
3. Exploring whether user engagement is associated with better health outcomes.

The analysis was done using three datasets: app events, user profiles, and health outcomes.

Data Overview

The dataset provided three tables.

App Events

This table contains timestamped records of actions users perform in the app, such as:

- starting the signup flow
- completing verification
- finishing profile setup
- opening the app
- recording health data

This table was used to understand user behaviour and activity patterns.

User Profiles

This table contains basic information about each user, including:

- signup date
- market (country)
- age band
- gender
- height
- subscription tier
- referral source

This dataset helped provide context about who the users are and when they joined the platform.

Health Outcomes

This table contains health measurements recorded over time, including:

- weight
- BMI
- body fat percentage

Some readings come from manual logs, while others come from connected devices or integrations.

Not all users have body fat data because this measurement requires a connected device.

Data Preparation and Quality Issues

Before beginning the analysis, the data was reviewed and cleaned. Several common issues were present in the dataset.

Duplicate Event Records

Some rows in the event dataset appeared more than once.

Exact duplicates were removed to avoid counting the same user action multiple times.

Inconsistent Text Values

Some fields contained inconsistent naming. For example, referral sources sometimes appeared with different formats (e.g., capitalization differences). These were standardized to ensure accurate grouping.

Missing Data

Not all users recorded health measurements. In particular, body fat percentage was only available for users with a connected device or integration.

For analyses involving body fat data, this limitation was noted and results were interpreted accordingly.

Timestamp Formatting

Date and time fields were converted into a consistent format so that user activity could be analysed over time.

Analysis

1. Activation Funnel Analysis

The first step was to examine the signup process.

Using the app events table, a funnel was built to track how many users move through each stage of the signup flow:

1. Signup flow started
2. Verification started
3. Verification completed
4. Profile setup completed
5. Signup completed

This helped identify where users drop out of the process.

The analysis showed that the **largest drop-off occurs between verification start and verification completion**. This suggests that the verification step may create friction for users during onboarding.

Defining “Fully Activated”

Signup completion alone does not necessarily mean a user has started using the product.

For this analysis, a user was considered **fully activated** when they completed signup **and performed at least one meaningful action**, such as:

- opening the app
- recording a health reading
- connecting a device

This definition reflects the idea that activation should represent the point where a user begins interacting with the product rather than simply creating an account.

2. Retention Analysis

Retention was used to understand how many users continue using the platform after signing up.

Users were grouped by signup week, and their activity was tracked in the following weeks.

A user was considered retained in Week 1 if they generated at least one app event during the week after signup.

The results showed that roughly half of new users return during their first week, although retention varies between signup cohorts.

This suggests that the first week of the user experience is an important period for engagement.

Behavioural Segmentation

To better understand user behaviour, users were grouped into segments based on their activity during the first 14 days after signup.

The following engagement groups were identified:

High Engagement

Users who interacted with the app frequently during their first two weeks.

Moderate Engagement

Users who showed some activity but used the app less consistently.

Low Engagement

Users who had little or no activity after signing up.

The analysis showed that users with higher early engagement were much more likely to return to the app later. This suggests that encouraging users to interact with the platform early in their journey could improve retention.

3. Health Outcomes Analysis

The health outcomes dataset was used to examine trends in weight and body fat percentage over time. Measurements were tracked for up to 12 weeks where available.

Because body fat percentage is only recorded through connected devices or integrations, analyses involving this metric were limited to users with available data.

Defining Meaningful Improvement

For this analysis, meaningful improvement was defined as a 5% reduction in weight compared to a user's starting measurement.

This threshold was chosen because small weight changes can occur naturally due to daily variation, while a 5% change is commonly used as a meaningful benchmark in health research.

Users who met this threshold were compared with those who did not to identify differences in behaviour and engagement.

4. AI Insight Quality Review

Several AI-generated health insights were reviewed to evaluate whether they are appropriate to send to users. Each message was assessed for accuracy, clarity, tone, and safety.

Insight A

Rating: Needs Edit

The message is positive but slightly misleading. While the blood pressure decrease is encouraging, the phrase “keep up whatever you're doing” is vague and could imply medical advice.

Improved version

Your blood pressure decreased from 145/92 to 138/88 this week, which is a positive change. Continuing healthy habits and following guidance from your healthcare provider may help support further improvement.

Insight B

Rating: Fail

A change of 0.4% in body fat is very small and may simply reflect normal measurement variation. Suggesting a possible metabolic issue could unnecessarily worry users.

Improved version

Your body fat percentage increased slightly over the past three weeks. Small changes can occur due to normal variation, so it may be helpful to continue tracking your measurements over time.

Insight C

Rating: Needs Edit

Comparing the user to “most users” may feel discouraging and does not provide helpful context.

Improved version

You logged 4 meals this week. Logging meals more consistently can help us provide more accurate insights about your nutrition patterns.

Insight D**Rating:** Pass

This message clearly explains the improvement and links it to specific behaviours such as increased steps and consistent sleep tracking. The tone is encouraging and informative.

Insight E**Rating:** Needs Edit

The message may feel slightly negative or discouraging when no data has been recorded.

Improved version

We didn't see any new data recorded this week. Logging your activity tomorrow can help keep your progress tracking up to date.

Key Assumptions and Limitations

Key Assumptions

Several assumptions were necessary to complete the analysis.

- A user was considered retained if they generated at least one app event during the week after signup.
- Engagement level was determined based on activity during the first 14 days after signup.
- Health improvement was defined as a 5% reduction in weight from the starting measurement.
- Device and manually logged health readings were assumed to be accurate.

These assumptions helped create consistent definitions for the analysis.

Limitations

There are several limitations that should be considered when interpreting the results.

First, the analysis identifies patterns in the data but does not prove cause and effect.

Second, users who connect devices or frequently log health data may already be more motivated than other users. This could influence both engagement levels and health outcomes.

Third, body fat data is only available for users with connected devices or integrations, which means those analyses represent only a subset of the user base.

Finally, the dataset covers a limited time period, so user behaviour may change as more data becomes available.

Recommendation

With more time or additional data, the analysis could be expanded in several ways.

First, it would be useful to explore how engagement patterns vary across markets, subscription tiers, or referral sources.

Second, a longer observation period would allow for deeper analysis of long-term retention and health outcomes.

Finally, additional behavioural signals such as education content usage or meal logging patterns could provide further insight into what drives user success.

Summary

The analysis highlights three key points.

First, the largest drop-off in the signup process occurs during the verification step, suggesting a potential area for product improvement.

Second, user engagement in the first two weeks after signup is strongly associated with whether users continue using the platform.

Third, some users show meaningful health improvement over time, though the available health data is limited to those who record measurements.

These findings suggest that improving the onboarding experience and encouraging early engagement may help increase both retention and long-term user outcomes.